

of the sample is the class with more class probability. Class '0' refers to the class of passengers who don't tip the driver and class '1' refers to the class of passengers who tip the driver. As can be seen from Figure 7, it is evident that according to the model, all three passengers won't offer a tip to the driver.

Results and discussion

In this article, we built an end-to-end pipeline to automate the workflow of deploying a machine learning model. The main advantage of the approach described in this project is that the entire pipeline can be described as a Python script. The output is a model running in a container that works as a server. It can hence be used by any other software application for making predictions by simply sending gRPC or JSON requests.

We have seen how ML models can be taken from the prototyping stage to deployment and then used in production. The field of MLOps is still young and there is a lot of research still going on.

This experimental work only explores a part of the automation of ML model

deployment. There are many other things that can be automated using DevOps. The pipeline can be made to retrain models as soon as new data is available or there is a drop in performance of the existing model. Also, many other orchestrators (such as KubeFlow) can be used in place of Apache Airflow. The article can also be taken forward

to include versioning of the data and model, as this is very important in the field of machine learning.

Note: The source code files used in this project are available at <https://github.com/mkd1997/taxi-pipeline-client>.



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